

Balaaditya Mukundan

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EDUCATION

University of California San Diego

Master of Science in Computer Science. GPA : 4.0/4.0

La Jolla, CA, USA

Sep 2024 – May 2026

College of Engineering Guindy, Anna University

Bachelor of Engineering in Computer Science. GPA : 4.0/4.0

Chennai, India

Aug 2020 – May 2024

TECHNICAL SKILLS

Languages: Python, Java, Go, C++, JavaScript, TypeScript

Systems & Backend: Spring Boot, Flask, FastAPI, Node.js, Ruby on Rails, Gin, GraphQL

Databases & Storage: PostgreSQL, MySQL, Redis, DynamoDB

Frontend: React, HTML, CSS, Tailwind

Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, CI/CD

EXPERIENCE

AquaMesh

April 2026 – Present

Software Development Engineer Intern

San Diego, CA

- Built backend tool-calling infrastructure for an AI chatbot over real-time aquatic sensor data using AWS Lambda, DynamoDB, and AppSync.
- Developed MCP-based serverless tools for time-series analytics, threshold monitoring, and anomaly detection on a multi-tenant IoT platform.
- Integrated AWS Amplify, Cognito, and Bedrock-based LLM workflows in a production environment.

Anna University

Jan. 2024 – May 2024

Generative AI Intern

Chennai, India

- Built a multimodal pipeline that converted STEM prompts into educational videos using fine-tuned LLM-generated Manim animations.
- Fine-tuned LLMs for mathematical animation code generation, improving syntactic correctness of generated Manim code and reducing generation errors by 48%.
- Integrated Tacotron2 speech synthesis and automated rendering workflows for end-to-end educational video generation.

Amazon

June 2023 – Aug 2023

Software Development Engineer Intern

Bangalore, India

- Reverse-engineered a legacy Ruby on Rails permission workflow reliant on manual SQL deployments across two codebases.
- Built a Spring Boot REST API layer and Rails-based self-service UI to automate permission provisioning, reducing turnaround from hours to minutes for **20+** engineers.
- Implemented 5 service-to-service APIs using SQL and DynamoDB, reducing operational overhead by **50%** across interconnected Rails and Spring Boot systems.
- Drove 80%+ test coverage using Mockito & JOOQ; authored LLD documents defining API contracts, data models, and failure-handling strategies for production deployment.

PROJECTS

API Abuse Protection Gateway | Go, Redis, PostgreSQL, Docker | [GitHub](#)

- Built a high-throughput multi-tenant API gateway in Go for concurrent request handling, abuse detection, and rate limiting.
- Designed a middleware-driven pipeline using Redis token bucket rate limiting with automatic API key and IP blocking.
- Sustained **6.7k req/sec** at **< 40ms** p95 latency under 200 concurrent clients across 20k requests on load test

Voice of Customer (VoC) Platform | Python, Redis, PostgreSQL, Streamlit | [GitHub](#)

- Built an event-driven platform to centralize multi-source customer feedback into a real-time analytics pipeline for teams.
- Designed asynchronous ingestion and processing pipelines using Redis Pub/Sub and PostgreSQL, ensuring reliable handling and persistence of **10K+ events/day** for search and analytics.
- Built a Streamlit analytics dashboard enabling product teams to query and surface trends across 10K+ daily feedback events in near real-time.

Azure Cloud-Native Intoxication Detection | Azure IoT Hub, Azure Functions, SQL, React | [GitHub](#)

- Built a real-time telemetry pipeline processing **1,000+ events/min** from distributed mobile sensors for live prediction.
- Designed an event-driven telemetry pipeline with asynchronous ingestion for bursty, high-frequency sensor streams.
- Developed a React dashboard for low-latency visualization of multi-device telemetry and prediction outputs.